Polar structure and D2 inference

From an explicit message matrix to structured multiuser decoding

8 October 2026 — derivation, related work, and Job 034

Question. Can we preserve the useful joint interference treatment of D2 without keeping one state per possible message? Polar coding gives a compact description of valid codewords, but does not by itself solve activity, pattern or multiuser inference. This note locates the overlap and the remaining approximation choices.

Status. Only the symbol-energy and shared-mixer encoder experiments are implemented here. Both retain the existing, full-message D2 recurrence. A new polar-based receiver is deliberately not selected or implemented. The evidence motivating this work is in the supervisor report; this note develops the algebra rather than repeating the result tables. Literature claims are cited; the structural identities below are derived for our model.

1 One signal model, two descriptions of the encoder

For K simultaneous transmissions in the real Gaussian MAC,

$$y = \sum_{k=1}^{K} \phi(w_k) + z = \Phi a^\star + z, \qquad z \sim \mathcal{N}(0, \sigma^2 I_n), \quad \Phi = [\phi(w)]_{w\in\{0,1\}^B} \in \mathbb{R}^{n\times M}, \quad M = 2^B. \tag{1}$$

Each column has energy one. The vector $a^\star \in \mathbb{Z}_{\geq 0}^M$ counts messages and sums to K. It is binary when messages are distinct. Our comparisons supply K and σ^2 to the receiver; with physical $E_b/N_0 = \eta$, $\sigma^2 = 1/(2B\eta)$. No unknown fading is present.

Split the message as $w = (p, v)$, with B_p pattern bits and $B_c = B - B_p$ payload bits. Let $P = 2^{B_p}$ and $V = 2^{B_c}$, so $M = PV$. The actual CRC and polar encoder are linear over binary arithmetic:

$$v \in \mathrm{GF}(2)^{B_c} \;\longmapsto\; c(v) = Lv \pmod 2 \in \mathrm{GF}(2)^{N_c} \;\longmapsto\; b(v) = (-1)^{c(v)} \in \{-1, +1\}^{N_c}. \tag{2}$$

Here L has shape $N_c \times B_c$ and includes CRC generation, insertion into unfrozen positions and the polar transform. Binary arithmetic describes the *code*, not the physical channel alphabet. The transmitted waveform is

$$\boxed{\phi(p, v) = A_p b(v)}, \qquad A_p \in \mathbb{R}^{n\times N_c}. \tag{3}$$

For ordinary ODMA, $A_p = S_p/\sqrt{N_c}$, where S_p inserts N_c symbols at distinct channel positions: $S_p^T S_p = I$. Our earlier notation $\phi = W_p x$, $x = b/\sqrt{N_c}$, is identical with $A_p = W_p/\sqrt{N_c}$. The matrices S_p specify placement; the polar code specifies which sign vectors are valid. They are separate choices.

1.1 What the XOR transform actually does

One kernel sends (u_0, u_1) to $(u_0 \oplus u_1, u_1)$, with $\oplus$ denoting addition modulo two. Applying this recursively gives the row-vector convention $c^T = u^T F^{\otimes q}$, $F = \left[\begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix}\right]$, $N_c = 2^q$. For example, $u = (0, 0, a, b)$ gives $c = (a \oplus b, b, a \oplus b, b)$; $(a, b) = (1, 0)$ gives signs $(-1, +1, -1, +1)$. The transform itself is invertible and adds no redundancy. Fixing input bits and imposing CRC constraints restricts the valid outputs to 2^{B_c} possibilities. Encoding costs $O(N_c \log N_c)$, without enumerating those possibilities. This is the structural starting point of polar coding [1]; our code uses a mother-code reliability ordering, not the full 5G NR transport chain.

2 D2: derive the implemented update

D2 represents uncertainty about *every complete message*, not individual polar bits. At iteration t, $a_j^t \in [0,1]$ estimates activity and $v_j^t = a_j^t(1-a_j^t)$ is its Bernoulli variance. Initially $a_j^0 = \rho = K/M$. Define

$$G = \Phi^T\Phi, \qquad d_j = G_{jj}, \qquad s_j = \sum_\ell G_{j\ell}^2. \tag{4}$$

Although useful in the derivation, the $M \times M$ matrix G is not built in the implementation.

2.1 Exact interference-cancellation identity

Form the residual $r^t = y - \Phi a^t$, then take its adjoint correlations $g^t = \Phi^T r^t$. Adding back the current coordinate's own contribution gives

$$\zeta_j^t = g_j^t + d_j a_j^t \tag{5}$$

$$= d_j a_j^\star + \sum_{\ell \neq j} G_{j\ell}(a_\ell^\star - a_\ell^t) + \phi_j^T z. \tag{6}$$

This identity is exact. The remaining term combines physical noise and errors in cancelling other messages. It motivates a scalar observation model $\zeta_j^t \approx d_j a_j^\star + \mathcal{N}(0, \tau_j^t)$.

2.2 Where approximation enters

If cancellation errors were uncorrelated, their interference variance would be $\sum_{\ell \neq j} G_{j\ell}^2 v_\ell^t$. D2 replaces these individual variances by their mean:

$$\bar{v}_{-j}^t = \frac{\sum_\ell v_\ell^t - v_j^t}{M-1}, \qquad \tau_j^t = \mathrm{softplus}(\beta_t)\left[\underbrace{\sigma^2 d_j}_{\text{physical noise}} + \underbrace{\bar{v}_{-j}^t(s_j - d_j^2)}_{\text{residual interference}}\right]. \tag{7}$$

Equal error variances would justify this averaging *within the uncorrelated-error model*. Iterative reuse of y and the fixed-K constraint generally violate independence. The learned scale calibrates the combined approximation; it does not separately identify physical noise and interference. There is no Onsager correction or state-evolution guarantee: this is not an AMP derivation.

2.3 Gaussian evidence, cardinality and damping

For Gaussian hypotheses with means d_j and 0, the log likelihood ratio is

$$e_j^t = \log \frac{p(\zeta_j^t \mid a_j^\star = 1)}{p(\zeta_j^t \mid a_j^\star = 0)} = \frac{d_j(\zeta_j^t - d_j/2)}{\tau_j^t}. \tag{8}$$

D2 then uses

$$q_j^t = \mathrm{sigmoid}\left(\log\frac{\rho}{1-\rho} + e_j^t + \lambda_t\right), \quad \sum_j q_j^t = K, \qquad a^{t+1} = \delta_t a^t + (1-\delta_t)q^t, \quad \delta_t = \mathrm{sigmoid}(\gamma_t). \tag{9}$$

The scalar λ_t enforces total *mean* activity. This is not the exact posterior obtained by conditioning independent Bernoulli variables on their sum. There are two learned scalars per layer, (β_t, γ_t): 16 for the eight-layer experiment. The final readout projects onto nonnegative integer counts summing to K.

Training backpropagates through the residual, geometry, scalar inference and encoder. The existing objective combines layerwise balanced activity BCE on e^t with a small count-regression term; it is not likelihood maximization. IID evaluation additionally tests a mismatch: repeated messages can have count greater than one, whereas D2's internal prior remains Bernoulli. None of these approximations makes D2 an optimal decoder.

3 Which matrix operations can polar structure replace exactly?

Collect the signs into $\mathcal{B} = [b(v)]_v \in \mathbb{R}^{N_c \times V}$. The explicit matrix is

$$\Phi = [A_0\mathcal{B} \mid A_1\mathcal{B} \mid \cdots \mid A_{P-1}\mathcal{B}]. \tag{10}$$

Partition $a = (a_0, \ldots, a_{P-1})$ with $a_p \in \mathbb{R}^V$. Then

$$\Phi a = \sum_p A_p m_p, \quad m_p = \mathcal{B} a_p \in \mathbb{R}^{N_c}, \qquad (\Phi^T r)_{p,v} = b(v)^T A_p^T r. \tag{11}$$

These are exact factorizations, not alternative decoders. The first needs only each pattern's mean symbol vector *after that mean has been computed*; the second turns message correlation into a score on a valid polar word. But arbitrary a_p still has V entries. A mean symbol vector does not determine the nonlinear next D2 update, its message-specific variance or its best messages. There is no exact closed update on m_p established here.

3.1 An exact shortcut for D2's geometry

Let ℓ_i^T be row i of the binary generator L. Over all equally weighted payloads,

$$\frac{1}{V}\sum_v b_i(v) b_j(v) = \frac{1}{V}\sum_v (-1)^{(\ell_i \oplus \ell_j)^T v} = \mathbf{1}\{\ell_i = \ell_j\} =: Q_{ij}. \tag{12}$$

If the two binary rows differ, pairing each v with a payload that flips their parity cancels the sum. If they agree, every summand is one. Thus $\mathcal{B}\mathcal{B}^T = VQ$ *exactly*. Repeated rows, including repeated zero rows, matter: assuming $Q = I$ would be wrong in general.

Define the small frame operator $\mathcal{F} = \Phi\Phi^T \in \mathbb{R}^{n \times n}$. Then

$$\boxed{\mathcal{F} = V \sum_p A_p Q A_p^T}, \qquad d_{p,v} = \|A_p b(v)\|^2, \qquad s_{p,v} = b(v)^T A_p^T \mathcal{F} A_p b(v). \tag{13}$$

This computes the frame without summing over all payloads or forming G. Job 034 uses this identity, with exact factorized forward/adjoint products. Tests compare both values *and gradients* against explicit multiplication and the original D2 geometry. The resulting D2 recurrence is unchanged, up to floating-point arithmetic.

3.2 A useful bridge, not a solved inverse problem

For a fixed pattern, maximizing the initial correlation is

$$\arg\max_{v \text{ valid}} b(v)^T t_p, \qquad t_p = A_p^T r. \tag{14}$$

This has the form of decoding a polar word from bit likelihoods. For white Gaussian noise and constant-energy codewords it is also the isolated-message ML score, with bit LLRs $2t_p/\sigma^2$. A bounded polar list could therefore propose messages without scanning all V. That is a credible computational bridge, not an exact replacement for D2: later D2 evidence contains message-dependent uncertainty and self-estimates; list pruning can omit true messages; unknown patterns and multiple messages on one pattern remain. A compact *atom generator* is not yet a compact *search algorithm*.

Scaling boundary. Per-message polar encoding is polynomial in N_c; storing a bank of P patterns is exponential only in B_p, not the full B. Our certification adapter still stores local payload signs and maintains $M = PV$ decoder states. It is intentionally restricted to small B. The geometry identity alone does not change this.

4 What a polar decoder supplies

Suppose a front end supplies likelihood information about the coded bits. Write its LLR as

$$\ell_i = \log \frac{p(\text{observation}_i \mid c_i = 0)}{p(\text{observation}_i \mid c_i = 1)}.$$

With conditionally independent bit observations and a uniform prior over valid codewords,

$$P(c \mid \ell) \;\propto\; \mathbf{1}\{c \text{ valid}\} \exp\left(\frac{1}{2}\sum_i \ell_i (-1)^{c_i}\right). \tag{15}$$

The validity indicator contains all the polar/frozen/CRC constraints. The expensive operation is summing over valid words to obtain marginals. Different polar decoders approximate or organize this inference differently.

4.1 Successive cancellation and lists

For the two-bit kernel, independent incoming LLRs combine as

$$f(\ell_1, \ell_2) = 2\,\mathrm{atanh}(\tanh(\ell_1/2)\tanh(\ell_2/2))\,, \tag{16}$$

$$g(\ell_1, \ell_2, \widehat{u}_0) = (-1)^{\widehat{u}_0}\ell_1 + \ell_2. \tag{17}$$

The f operation infers the XOR bit by marginalizing the other bit. Once that XOR is hypothesized, g aligns the two pieces of evidence for the second bit. Recursive application gives successive cancellation (SC): frozen bits are fixed, information bits are decided sequentially. A wrong early decision can propagate.

SCL branches on both information-bit values and retains a bounded number Q_{list} of paths, ranked by their likelihood metrics [2]. CRC filters completed candidates. Standard efficient implementations have complexity $O(Q_{\text{list}} N_c \log N_c)$ rather than $O(2^{B_c})$. A finite list is not exact ML, and passing CRC does not prove correctness. Our existing native receiver uses a hard CRC-aided list; it is not a soft-output joint receiver.

4.2 Soft output and the information sent back

A soft-output decoder estimates coded-bit posterior LLRs

$$\ell_i^{\text{APP}} = \log \frac{\sum_{c \text{ valid}: c_i = 0} P(c \mid \ell)}{\sum_{c \text{ valid}: c_i = 1} P(c \mid \ell)}. \tag{18}$$

Polar BP exchanges messages on the XOR graph. Alternatively, soft-output SCL approximates these sums from a list. Normalizing only over surviving paths can be severely overconfident when the correct path was pruned. Yuan et al. explicitly estimate the unvisited probability mass [6]; this remains an approximation, not an exact posterior or a guarantee of list coverage.

In an iterative detector/decoder loop, return *extrinsic* information:

$$\ell_i^{\text{ext}} = \ell_i^{\text{APP}} - \ell_i^{\text{in}}. \tag{19}$$

Here ℓ_i^{in} must include exactly the incoming evidence already supplied by the detector, with any other priors accounted for consistently. Returning the full APP as independent new evidence would repeatedly count the same observation. Extrinsic subtraction does not make a loopy graph exact, but specifies the intended separation.

What polar structure buys. It replaces an arbitrary prior on 2^{B_c} complete payloads by local algebraic constraints and bounded recursive inference. It does not turn a superposition of several unknown codewords into independent bit channels. Producing credible ℓ_i from that superposition is the multiuser detector's job.

5 Where compressed sensing and polar inference meet

An exact posterior can be written using K exchangeable message slots, not device identities:

$$P(\{p_k, v_k\}_{k=1}^{K} \mid y) \;\propto\; \exp\left[-\frac{\|y - \sum_k A_{p_k} b(v_k)\|^2}{2\sigma^2}\right] \pi(\{p_k, v_k\}). \tag{20}$$

The prior π must match distinct or IID sampling. Equivalently, expose the coded bits and impose the local polar, frozen and CRC constraints. Both descriptions represent the same channel. A factor graph reorganizes the problem; its channel-sum factors still couple users, so it is not a tractability proof.

5.1 Derive a soft interference step, conditional on patterns

For clarity, first suppose the K patterns are known, and stack their coded signs into b: $y = Hb + z$, $H = [A_{p_1} \mid \cdots \mid A_{p_K}] \in \mathbb{R}^{n \times KN_c}$. Index a sign by e. Incoming extrinsic LLRs give $m_e = \tanh(\ell_e^{\text{ext}}/2)$ and $v_e = 1 - m_e^2$. A diagonal, Gaussian interference approximation for resource i is

$$r_{i\to e} = y_i - \sum_{f \neq e} H_{if} m_f, \qquad T_{i\to e} = \sigma^2 + \sum_{f \neq e} H_{if}^2 v_f. \tag{21}$$

The two hypotheses now have means $+H_{ie}$ and $-H_{ie}$, hence

$$\boxed{\ell_{i\to e} = \frac{2H_{ie} r_{i\to e}}{T_{i\to e}}}. \tag{22}$$

Under a factorized resource approximation, sum these messages into each sign's detector LLR and send them to its polar decoder. Return extrinsic means and uncertainties, then repeat. This is the same basic cancellation/effective noise/Bayesian-update logic as D2, but the latent variables are coded signs rather than complete-message activities.

A vector version keeps colored interference: $R_{\to e} = \sigma^2 I + \sum_{f \neq e} v_f h_f h_f^T$, $r_{\to e} = y - \sum_{f \neq e} h_f m_f$, giving $\ell_{\to e} = 2h_e^T R_{\to e}^{-1} r_{\to e}$. This motivates soft LMMSE front ends. Both versions still approximate correlated code-bit uncertainty and the effect of reusing observations. Correlation-aware updates cost more; adding a learned scale does not make either exact.

	Full-message D2	Structured detector + polar inference
Latent state	M message activities	Pattern/activity state and coded-bit messages
Linear evidence	$\Phi^T(y - \Phi a)$	Placement/spreading residuals and bit LLRs
Prior step	Bernoulli + mean cardinality	Polar/CRC constraints + activity model
Main limitation	Exponential state; approximate uncertainty	Unknown-pattern search, collisions, loopy uncertainty

The unresolved part is substantial. Known patterns above are an explanatory conditioning, not an allowed oracle in the final receiver. Different full messages can share a pattern. With IID prefixes, a tagged user's pattern collides with probability $1 - (1 - 1/P)^{K-1}$, even when full-message collisions are negligible. A one-packet-per-pattern model would therefore change the problem. Activity/pattern inference must allow multiplicity, or a revisable list must represent competing messages and measure its misses. Simply replacing D2's sigmoid by SC does neither.

6 Existing work already covers important parts of this overlap

This is a focused map of relevant primary papers, not an exhaustive survey or a claim that their assumptions match Job 033. The opportunity must be narrower than "combine polar coding with interference cancellation".

Polar coding plus spreading in URA. Pradhan et al. [3] split the message into a spreading signature and a polar-coded payload. Their receiver detects signatures, forms LLRs with MMSE estimation, applies CRC-aided SCL, and cancels recovered packets. Thus polar coding without ODMA is already an established construction. It is a natural architectural reference, not a new idea in itself. Their large-payload operating point is not our short-message certification regime.

A soft-in/soft-out multiuser loop in URA. Pradhan et al. [4] replace the polar/list stage of that spreading architecture by LDPC BP and exchange soft information with an MMSE detector. This directly addresses the value of revisable uncertainty instead of only hard cancellation. It also shows that the code and the iterative detector can be designed together; polar is not the only possible constraint engine.

Polar BP with soft interference cancellation. Ebada et al. [5] study iterative polar decoding in two- and four-user Gaussian MACs using an IDMA construction. This is a close reference for the bit-level loop in Section 5, but it does not solve unsourced pattern/activity search. Its complexity claims should not be imported as a complete URA decoder guarantee.

Soft output from a strong list decoder. Yuan, Duffy and Médard [6] supply a route to calibrated soft information from SCL rather than requiring polar BP. This is relevant if plain BP gives up too much decoding quality. Their treatment of probability outside the list is particularly pertinent to candidate-based inference.

Learning the polar decoder, not only the code. Xu et al. [7] learn weighted message updates on a sparse polar decoding graph. Learned decoding of fixed polar constraints is therefore established prior art. A project-specific question would be whether learned multiuser uncertainty exchange helps, under matched waveforms, activity assumptions and computational budgets; it is not merely whether BP weights can be trained.

ODMA and soft polar feedback already meet. Li et al. [8] develop a MIMO URA architecture with pilots, hierarchical pattern detection and soft polar feedback. It is directly relevant to joint pattern/data inference. Its multi-antenna channel estimation and pilot structure differ from our real, known-channel MAC, so it is neither an interchangeable baseline nor evidence that our remaining inverse problem has already been solved.

6.1 What seems worth deriving next

The most informative comparison is between two uncertainty representations on *the same* fixed encoder:

1. A revisable polar-generated codeword list with joint residual refinement. Separate proposal recall and list size from the error of refinement; do not supply true messages or hide misses.
2. A soft detector/code loop, with explicit activity and same-pattern collision handling. Compare scalar interference variances with a colored/LMMSE alternative before adding many learned parameters.

Use known-pattern tests only to diagnose the inference core, then remove that oracle. Keep known K initially for comparison with D2, but label it; estimating K and controlling false alarms remain separate requirements.

Research position. D2's gain suggests useful uncertainty is lost by the current native receiver, not that D2's equations are uniquely correct. A tractable decoder may improve on D2 by using code constraints more faithfully. Conversely, better single-user polar decoding alone may not repair poor pattern evidence or an overconfident front end. No receiver choice is made by the encoder tests on the next page.

7 Job 034: two controlled encoder investigations

Keep the selected Job 033 polar/CRC code, pattern placements, seeds, observations and D2 training contract. Vary one part of A_p at a time. The unchanged ODMA–polar/D2 rows from 033 supply the fixed baseline; dense/sparse D2 and native ODMA rows from 032 are reused rather than retrained.

7.1 Investigation 2: learn symbol-energy profiles

For each pattern, learn $\theta_p \in \mathbb{R}^{N_c}$ and set

$$e_{p,i} = \frac{\exp \theta_{p,i}}{\sum_j \exp \theta_{p,j}}, \quad A_p = S_p \operatorname{diag}(\sqrt{e_p}), \qquad \|\phi(p,v)\|^2 = \sum_i e_{p,i} = 1 \quad \text{for every } v. \tag{23}$$

Initialize $\theta = 0$ to recover equal energy exactly. Polar bits, CRC and positions are fixed; the continuous energy logits and D2 parameters are trained jointly. This changes the relative importance of coded symbols and may change within-pattern distances. It is a structured PN_c-parameter family, not an arbitrary amplitude per message. The normalization is enforced in every forward call, including for previously unseen payloads.

7.2 Investigation 3: learn one shared pre-placement mixer

Learn a skew-symmetric matrix K_0 through its $N_c(N_c-1)/2$ upper-triangular entries and use the Cayley map

$$U = (I - K_0)^{-1}(I + K_0), \quad U^T U = I, \qquad A_p = \frac{1}{\sqrt{N_c}} S_p U. \tag{24}$$

Initialize $K_0 = 0$, giving $U = I$ and exactly the original codebook. This finite chart covers orthogonal matrices without eigenvalue -1 in the identity component; it is not an unrestricted real mixer. It preserves unit energy and every *within-pattern* distance, while changing cross-pattern interactions:

$$\phi(p,v)^T \phi(q,v') = \frac{1}{N_c} b(v)^T U^T S_p^T S_q U b(v'). \tag{25}$$

It is not a common rotation of the entire received waveform. An isolated known-pattern signal can be unmixed by $U^T S_p^T$, but multiuser interference becomes mixed too. Dense U adds $O(N_c^2)$ per-message work and can complicate future bit-level inference. Better exhaustive D2 performance would not automatically imply a practical receiver gain.

7.3 Contract, checks and interpretation

- Twelve rows: two families $\times$ $B = 12, 14$ $\times$ three paired seeds; $n = 256$. Selected $N_c = 32$, $B_p = 6, 8$: energy has 2,048/8,192 parameters; the shared mixer has 496.
- Eight-layer D2, 200 epochs maximum, patience 10 and restoration of the best encoder/decoder pair. Same loads $7, 15, 22, 26$, physical $E_b/N_0 = -4{:}2{:}10$ dB evaluation, distinct-message training and distinct/IID evaluation as 033. No energy-plus-mixer combined arm and no new receiver.
- Exact initialization matches all six returned 033 codebooks. Forward/adjoint products, geometry and gradients match explicit references; all generated codewords retain unit energy. A sampled $B = 100$ forward check tests generation only. The small-B D2 adapter explicitly rejects that decoding claim.

Local $B = 8$ mini-runs reduced validation loss from about 0.1082 to 0.0821 (fixed), 0.0758 (energy), and 0.0741 (mixer). These are execution and initial-learning checks, not production rankings; lower surrogate loss need not give lower PUPE in a tiny test. HPC results are pending. Compare paired per-load/SNR gaps against fixed polar D2 and dense D2: a gain identifies useful waveform freedom; no gain at this budget is not a proof of optimality.

References and implementation pointers

The derivation follows the real, unit-energy comparison system. It does not import fading, pilot or antenna gains from papers using different models. Report equations were checked against the retained implementations.

References

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[2] I. Tal and A. Vardy, *List Decoding of Polar Codes*, 2015 (preprint 2012). arXiv:1206.0050.

[3] A. K. Pradhan, V. K. Amalladinne, K. R. Narayanan and J.-F. Chamberland, *Polar Coding and Random Spreading for Unsourced Multiple Access*, 2019 preprint / ICC 2020. arXiv:1911.01009.

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[6] P. Yuan, K. R. Duffy and M. Médard, *Soft-Output Successive Cancellation List Decoding*, IEEE Transactions on Information Theory, 71(2), 1007–1017, 2025. arXiv:2412.03623.

[7] W. Xu, X. You, C. Zhang and Y. Be'ery, *Polar Decoding on Sparse Graphs with Deep Learning*, 2018. arXiv:1811.09801.

[8] T. Li et al., *ODMA-based MIMO Massive Unsourced Random Access with Soft-Output Polar Codes*, 2026 preprint; listed as accepted in IEEE Transactions on Wireless Communications. arXiv:2608.28085.

Implemented sources.

- `framework/learned_decoders.py`: unchanged D2 recurrence and its approximations.
- `baselines/polar.py`, `baselines/odma_polar.py`: binary code, placement and native list/SIC receiver.
- `framework/polar_mapping.py`: procedural mappings, small-B adapter and exact geometry shortcut.
- `benchmarks/polar_mapping.py`: unchanged training/evaluation contract via the shared benchmark.
- `jobs/034_polar_mapping/README.md`: manifest, controls, commands and local checks.
- `tests/polar_mapping_test.py`: algebra/gradient regressions; `tests/polar_mapping_learning.py`: smoke and mini runs.